

GG migration and cutover to RHEL 7 runbook

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- [Pre-requisites](#)
- [pre-checks](#)
- [Cutover steps to RHEL 7](#)
- [Post Cutover](#)
 - [Verify GG sync and ensure no lag between RHEL7 and RHEL6](#)
 - [AWS Steps](#)
- [Rollback](#)
 - [Post Rollback](#)
 - [AWS steps](#)

Pre-requisites

- create new pump on RHEL7 for AWS — [Completed](#)
 - create pump PPSPPP01 to pump trail files from RHEL7 to GG hub (RDS) .
- Switchover RHEL7 to LVDC

pre-checks

- Verify latest veridata compare run to make sure there are NO OOS rows – [Completed](#)
- If its there we need to fix it first before cutover. Below is example : [– NA](#)

1. log in to 12c db as GGT user
2. run 15 days veridata compare check OOS rows in veridata [for](#) table and OOS rows and columns
3. run below update on target(12c) database as GGT user. Change table name as per OOS report.

- Verify GG replication status on both sides

1.

```
GGSCI (pprdpspdb900.ie.intuit.net) 1> info all
```

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EPSPLPD1	00:00:00	00:00:04
EXTRACT	RUNNING	PPSPLPD2	00:00:01	00:00:02
REPLICAT	RUNNING	RPSPLPD2	00:00:00	00:00:07

```
GGSCI (pprdpspdb901.ie.intuit.net) 1> info all
```

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EPSPLPD2	00:00:04	00:00:06
EXTRACT	RUNNING	PPSPLPD1	00:00:00	00:00:06
REPLICAT	RUNNING	RPSPLPD1	00:00:00	00:00:07

2.

Check lag via heartbeat as well as below:

```
login to 12c (pprdpspdb901.ie.intuit.net)
```

```
ALTER SESSION SET NLS_DATE_FORMAT='mm-dd-yyyy hh24:mi:ss';
select * from pspadm.gg_heartbeat;
```

- stop service and Lock application user accounts on current database (RHEL 6) .

1.

Login to pprdpspdb900.ie.intuit.net and stop prod service

```
SQL> exec dbms_service.stop_service('pspprod');
```

PL/SQL procedure successfully completed.

login to pprdpspdb800.ie.intuit.net and stop DR service

```
SQL> exec dbms_service.stop_service('pspdr');
```

PL/SQL procedure successfully completed.

2.

```
alter user PSPADM account lock;
alter user PSPAPP account lock;
alter user PSPBOADMIN account lock;
alter user PSPFLUXADM account lock;
alter user PSPLOG account lock;
alter user PSPFLUXADMIN08 account lock;
alter user PSPREAD account lock;
alter user PSPTEMP account lock;
alter user APPBUGFIX account lock;
alter user PSPREPORTS account lock;
alter user PSPHIST_QBOE account lock;
alter user ARCADM account lock;
alter user EBIETL account lock;
alter user ICADMIN account lock;
```

Kill existing connections (**if** still exist)

```
select 'alter system kill session '||''''||sid||','||serial#||''''||' immediate;' from gv$session where username in
```

```

      USERNAME                                ACCOUNT_STATUS
-----
ARCADM                                LOCKED
PSPHIST_QBOE                          LOCKED
APPBUGFIX                             LOCKED
PSPLOG                                LOCKED
PSPFLUXADM                            LOCKED
PSPFLUXADMIN08                        LOCKED
PSPADM                                LOCKED
PSPTEMP                               LOCKED
PSPBOADMIN                            LOCKED
PSPAPP                                LOCKED
PSPREPORTS                            LOCKED
EBIETL                                LOCKED
PSPREAD                               LOCKED
ICADMIN                               LOCKED

14 rows selected.
```

- Verify GG replication

```

1.
GGSCI (pprdpspdb900.ie.intuit.net) 1> info all

Program      Status      Group      Lag at Chkpt  Time Since Chkpt
-----
MANAGER      RUNNING
EXTRACT      RUNNING     EPSPLPD1   00:00:00      00:00:04
EXTRACT      RUNNING     PPSPLPD2   00:00:01      00:00:02
REPLICAT     RUNNING     RPSPLPD2   00:00:00      00:00:07

GGSCI (pprdpspdb901.ie.intuit.net) 1> info all
```

MANAGER	RUNNING			
EXTRACT	RUNNING	EPSPLPD2	00:00:04	00:00:06
EXTRACT	RUNNING	PPSPLPD1	00:00:00	00:00:06
REPLICAT	RUNNING	RPSPLPD1	00:00:00	00:00:07

2. Check lag via heartbeat as well as below:
login to 12c (pprdpspdb901.ie.intuit.net)

```
ALTER SESSION SET NLS_DATE_FORMAT='mm-dd-yyyy hh24:mi:ss';
select * from pspadm.gg_heartbeat;
```

- restart extract EPSPLPD2 on RHEL6
- check and verify sequence between RHEL6 and RHEL7

```
set pagesize 1000
set numwidth 30
```

```
col SEQUENCE_OWNER format a10
col SEQUENCE_NAME format a32
```

```
select SEQUENCE_OWNER, SEQUENCE_NAME, LAST_NUMBER from dba_sequences where SEQUENCE_OWNER in ('PSPADM') order by SE
```

- export and import few db schemas from db901 (RHEL6) to db800 (RHEL 7)

1. Take export from db server db901

```
create or replace directory DATA_PUMP_DIR as '/u01/ogg/exp_bkp';
```

```
nohup expdp userid='/' as sysdba\ ' schemas=PSPLOG,PSPTMP,APPBUGFIX,PSPFLUXADM directory=DATA_PUMP_DIR dumpfile=sch
```

2. scp to db server db800

```
Drop user PSPTEMP cascade;
Drop user APPBUGFIX cascade;
Drop user PSPFLUXADM cascade;
```

4. `import`

```
nohup impdp userid='/' as sysdba\ ' schemas=PSPLOG,PSPTEMP,APPBUGFIX,PSPFLUXADM directory=DATA_PUMP_DIR dumpfile=sch
```

- Unlock application user accounts on database (RHEL 7)

1.

```
alter user PSPADM account unlock;
alter user PSPAPP account unlock;
alter user PSPBOADMIN account unlock;
alter user PSPFLUXADM account unlock;
alter user PSPLOG account unlock;
alter user PSPFLUXADMIN08 account unlock;
alter user PSPLOG account unlock;
alter user PSPREAD account unlock;
alter user PSPTEMP account unlock;
alter user APPBUGFIX account unlock;
alter user ARCADM account unlock;
alter user EBIETL account unlock;
alter user ICADMIN account unlock;
```

2. verify

```
select username,account_status from dba_users where username like '%PSP%' or username in ('APPBUGFIX','ARCADM','ICA
USERNAME ACCOUNT_STATUS
```

```
-----
ARCADM OPEN
PSPHIST_QBOE OPEN
APPBUGFIX OPEN
```

PSPADM	OPEN
PSPTEMP	OPEN
PSPBOADMIN	OPEN
PSPAPP	OPEN
PSPREPORTS	OPEN
EBIETL	OPEN
PSPREAD	OPEN
ICADMIN	OPEN

14 rows selected.

Enable sequence replication between RHEL 7 and RHEL 6 (one time step)

```
SQL> @sequence.sql
Please enter the name of a schema for the GoldenGate database objects:
ggs
Setting schema name to GGS

UPDATE_SEQUENCE STATUS:

Line/pos
-----
Error
-----

No errors
No errors

GETSEQFLUSH

Line/pos
-----
Error
-----
```

```
SEQTRACE
```

```
Line/pos
```

```
-----
```

```
Error
```

```
-----
```

```
No errors
```

```
No errors
```

```
REPLICATE_SEQUENCE STATUS:
```

```
Line/pos
```

```
-----
```

```
Error
```

```
-----
```

```
No errors
```

```
No errors
```

```
STATUS OF SEQUENCE SUPPORT
```

```
-----
```

```
SUCCESSFUL installation of Oracle Sequence Replication support
```

Post Cutover

Verify GG sync and ensure no lag between RHEL7 and RHEL6

EXTRACT EQBQPD16 Last Started 2017-11-19 21:24 Status STOPPED
Checkpoint Lag 00:00:02 (updated 00:00:24 ago)
Log Read Checkpoint Oracle Integrated Redo Logs
2017-11-28 02:52:22
SCN 1976.1787450233 (8488642827129)

Target Extract Trails:

Trail Name	Seqno	RBA	Max MB	Trail Type
./dirdat/qboqpd16/tr	4692	119910231	500	EXTTRAIL

Integrated Extract outbound server first scn: 1976.788177432 (8487643554328)

Integrated Extract outbound server filtering start scn: 1976.788177432 (8487643554328)

Extract Source	Begin	End
Not Available	2017-05-24 20:54	2017-11-28 02:52
Not Available	2017-05-24 20:54	2017-11-19 21:21
Not Available	2017-05-24 20:54	2017-10-26 22:38
Not Available	2017-05-24 20:54	2017-09-26 15:02
Not Available	2017-05-24 20:54	2017-08-03 22:41
Not Available	2017-05-24 20:54	2017-07-13 23:12
Not Available	2017-05-24 20:54	2017-06-09 20:24
Not Available	2017-05-24 20:54	2017-06-09 04:55
Not Available	2017-05-24 20:54	2017-06-08 22:47
Not Available	* Initialized *	2017-05-24 20:54
Not Available	* Initialized *	2017-05-24 20:54

Current directory: /u01/cgg/qboqpd16/12_1_2_1_6

```
Checkpoint file      /u01/ogg/qboqpd16/12.1.2.1.6/dirchk/EQBQPD16.cpe
Process file        /u01/ogg/qboqpd16/12.1.2.1.6/dirpcs/EQBQPD16.pce
Error log           /u01/ogg/qboqpd16/12.1.2.1.6/ggserr.log
```

AWS Steps

- **take a snapshot backup** for database in question - psppp001
- stop pump PPITPP01 on RHEL6
- Verify replicat (from RHEL6) status is at EOF on GG hub

```
send status [extract / replicat]
```

- Create new replicat RPSPQPD4 on GG hub for RDS
- stop replicat from RHEL6 and rename parameter file to dummy on GG hub
- start pump PPITPP01 on RHEL6
- Verify OGG monitoring to db800, db901, GG hub server for all GG processes.
- Total proceses on db800 in running status (RHEL7)
 - 1 extract ~~for RHEL6 and AWS~~
 - 2 pump for RHEL6 and AWS
 - 1 replicat ~~from RHEL6~~
- Total processes on db901 in running status (RHEL6)
 - 1 extract for monolith
 - 2 extracts for IBOB
 - 1 pump for RHEL7
 - 1 pump for AWS
 - 1 replicat for monolith
 - 2 replicats for IBOB
- On AWS in running status
 - 1 replicat for RHEL 7
 - 1 STOPPED replicat for RHEL6
- Test GG replication by insert/update/delete on test table under PSPADM schema

- stop application, lock accounts and kill existing session (if required after app shutdown) on RHEL7
- verify GG sync
- restart extract EPSPQPD4 on RHEL 7
- import/export schemas back to RHEL6
- check and verify sequence replication between RHEL6 and RHEL7

```
set pagesize 1000
set numwidth 30

col SEQUENCE_OWNER format a10
col SEQUENCE_NAME format a32

select SEQUENCE_OWNER, SEQUENCE_NAME, LAST_NUMBER from dba_sequences where SEQUENCE_OWNER in ('PSPADM') order by SE
```

- Unlock users and start application

Post Rollback

AWS steps

- stop pump PPSPPP01 to AWS on RHEL7
- verify replicat from RHEL7 status is at EOF on GG hub
- stop replicat RPSPQPD4 on GG hub and rename parameter file to dummy
- start old replicat RPSPLPD2 on GG hub
- ~~start old pump PPITPP01 on RHEL6~~
- Test GG replication by insert/update/delete on test table under PSPADM schema

No labels